



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PALANISAMY AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

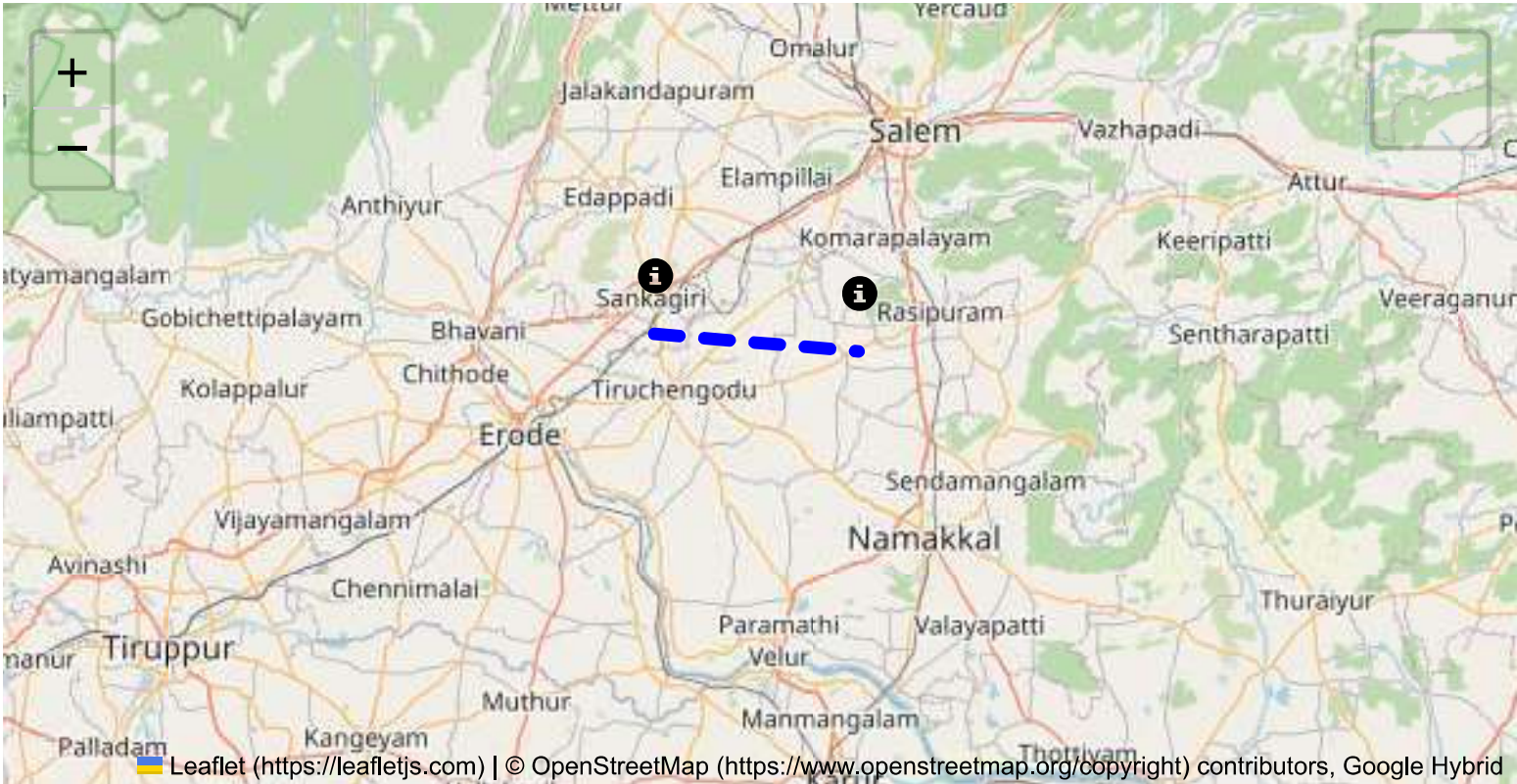
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

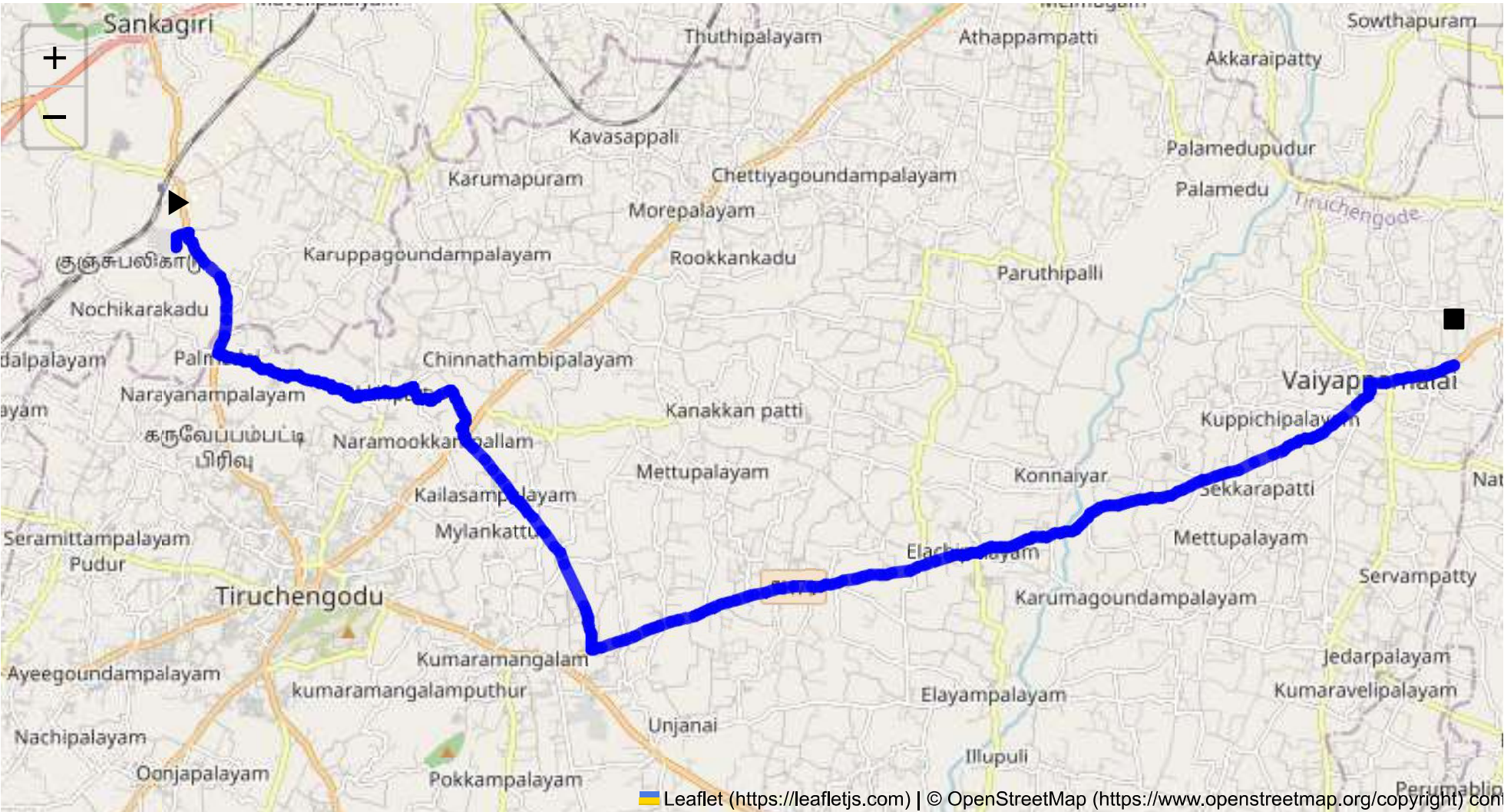
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 31.34 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.418155, 78.095109)



Welcome to the Journey Risk Management Study

Comprehensive Route Safety Analysis

1. Overview of the Route Map

The route from Sangagiri to Vaiyappamalai predominantly follows NH44. This highway is a national route that provides direct connectivity between the two points, passing through semi-rural and urban areas.

The total distance is approximately 31.34 kilometers.

2. Typical Weather Conditions and Potential Hazards

The region experiences a tropical climate with the most significant weather challenges occurring during the monsoon season (June to September), when heavy rains can lead to flooding and poor visibility. During the summer months (March to May), high temperatures can affect vehicle performance, particularly cooling systems.

3. Traffic Patterns

- **Peak Hours:** Typically, the peak traffic hours are observed during morning (8:00 AM to 10:00 AM) and evening (5:00 PM to 7:00 PM).
- **Congestion-Prone Areas:** The stretch near urban centers and market areas, particularly around Sangagiri town, may experience higher congestion. Traffic slows considerably near toll booths and road junctions.

4. Assessment of Road Quality and Infrastructure

NH44 is generally well-maintained, but some sections may show signs of wear, including potholes or uneven surfaces, especially post-monsoon. Watch for construction zones which might not be well-demarcated.

5. Alternative Routes for Emergencies

In the event of a route blockage on NH44, an alternative route via SH86 through Tiruchengodu provides a viable option, though it adds approximately 15 to 20 kilometers to the journey.

6. Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials through this route is subject to specific regulations requiring permits. Restrictions are stricter for carrying highly inflammable or toxic substances, with local enforcement agencies conducting regular checks.

7. Overview of Historical Incidents

Historically, there have been few major incidents involving heavy vehicles on this route. However, minor accidents occur more frequently during peak hours, often due to human error or adverse weather conditions.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural lands and small residential zones. It is crucial to minimize emissions and noise to avoid disturbing these areas. Proper spill response protocols should be in place to protect the local environment in the event of an accidental discharge of hazardous materials.

9. Communication Coverage

Most of the NH44 route is covered by mobile networks, offering reliable communications. However, there are potential dead zones near less populated regions, so it is advised to prepare for limited connectivity in such areas.

10. Estimated Emergency Response Times

Emergency services have varying response times along the route. In urban and semi-urban areas, response times are typically around 20-30 minutes. In more rural stretches, it might extend up to 45 minutes to an hour due to distance and road access.

12. Overall Risk Assessment Summary

The overall route risk is moderate, with the primary risks arising from weather conditions, road wear, and traffic congestion during peak hours. Proper vehicle maintenance, awareness of alternative routes, and compliance with local regulations will mitigate most risks associated with transporting hazardous materials.

In summary, this route is feasible and generally safe for transporting hazardous materials, provided that all safety protocols and regulations are strictly followed and contingencies are planned for adverse weather and road conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41975, 77.88075	10 KM/Hr
5	Turn	Medium	11.41876, 77.88487	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	High	11.41859, 77.88692	15 KM/Hr
8	Turn	Medium	11.41852, 77.88700	30 KM/Hr
9	Turn	Medium	11.41769, 77.88730	30 KM/Hr
10	Turn	Medium	11.41630, 77.89077	30 KM/Hr
11	Turn	Medium	11.41601, 77.89198	30 KM/Hr
12	Turn	Medium	11.41603, 77.89479	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.41439, 77.90025	30 KM/Hr
14	Turn	Medium	11.41377, 77.90073	30 KM/Hr
15	Turn	Medium	11.41389, 77.90161	30 KM/Hr
16	Turn	Medium	11.41325, 77.90313	30 KM/Hr
17	Turn	Medium	11.41276, 77.90349	30 KM/Hr
18	Turn	Medium	11.41227, 77.90470	30 KM/Hr
19	Turn	Medium	11.41235, 77.90492	30 KM/Hr
20	Turn	Medium	11.41271, 77.90516	30 KM/Hr
21	Turn	High	11.41329, 77.90978	15 KM/Hr
22	Turn	Medium	11.41314, 77.91006	30 KM/Hr
23	Turn	High	11.41313, 77.91057	15 KM/Hr
24	Turn	High	11.41307, 77.91058	15 KM/Hr
25	Turn	Medium	11.41334, 77.91266	30 KM/Hr
26	Turn	High	11.41424, 77.91433	15 KM/Hr
27	Turn	Medium	11.41409, 77.91458	30 KM/Hr
28	Turn	High	11.41208, 77.91551	15 KM/Hr
29	Turn	Medium	11.41207, 77.91604	30 KM/Hr
30	Turn	High	11.41204, 77.91610	15 KM/Hr
31	Turn	Medium	11.41216, 77.91626	30 KM/Hr
32	Turn	Medium	11.41183, 77.91755	30 KM/Hr
33	Turn	Medium	11.41342, 77.92087	30 KM/Hr
34	Blind Spot	Blind Spot	11.41350, 77.92093	10 KM/Hr
35	Turn	Medium	11.41347, 77.92096	30 KM/Hr
36	Turn	Medium	11.41336, 77.92130	30 KM/Hr
37	Turn	High	11.40566, 77.92355	15 KM/Hr
38	Turn	High	11.40557, 77.92343	15 KM/Hr
39	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
40	Turn	Medium	11.41463, 78.08067	30 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41975, 77.88075



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41829, 77.88568



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41859, 77.88692



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41852, 77.88700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41769, 77.88730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41630, 77.89077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41601, 77.89198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41377, 77.90073



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41389, 77.90161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41325, 77.90313



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41227, 77.90470



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41235, 77.90492



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41271, 77.90516



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41314, 77.91006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41313, 77.91057



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41307, 77.91058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41334, 77.91266



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41207, 77.91604



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41204, 77.91610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41216, 77.91626



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41183, 77.91755



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41342, 77.92087



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41350, 77.92093



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41463, 78.08067